

Phasor

AC



V, I



Sinusoid

$$R = \frac{V}{I}$$

$$Z = \frac{V}{I}$$

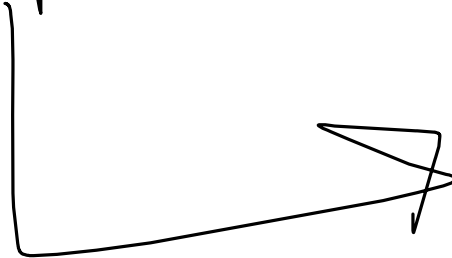
$$I = A_1 \sin(\omega t + \phi_1)$$

$$V = A_2 \sin(\omega t + \phi_2)$$

Phasor



complex number



$$2 \angle 30^\circ$$

✓ $v = 2 \cos[5t + 30^\circ]$

Complex number - 3 form

z can be represented in three ways.

$$\left[\begin{array}{ll} z = x + jy & \text{Rectangular form} \\ z = r \angle \phi & \text{Polar form} \\ z = re^{j\phi} & \text{Exponential form} \end{array} \right. \quad (9.15)$$

$$z = x + jy \quad r = \sqrt{x^2 + y^2}$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$x = r \cos \phi \quad y = r \sin \phi$$

$$z = r e^{j\phi} = r (\cos \phi + j \sin \phi)$$

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2) \quad (9.18b)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2 \quad (9.18c)$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2 \quad (9.18d)$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \angle -\phi \quad (9.18e)$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \phi/2 \quad (9.18f)$$

Complex Conjugate:

$$z^* = x - jy = r \angle -\phi = re^{-j\phi} \quad (9.18g)$$

Note that from Eq. (9.18e),

$$\frac{1}{j} = -j \quad (9.18h)$$

Handwritten notes:

$z^* = r \angle -\phi$

$\frac{1}{j} = -j$

Phonons

$$V(t) = V_m \cos(\omega t + \phi)$$

time domain
10 um case.

$$V = V_m \cos \phi$$

phonon domain
10 um case.

$$\frac{dv}{dt} \rightarrow j\omega V$$

$$\int v dt \rightarrow \frac{1}{j\omega} X V$$

Express these sinusoids as phasors:

(a) $v = 7 \cos(2t + 40^\circ)$ V ✓

(b) $i = -4 \sin(10t + 10^\circ)$ A

1a) $v = 7 \cos(2t + 40^\circ)$
 $\underline{v} = 7 \angle 40^\circ$

$$\cancel{v(t)} = \cancel{V_m} \cos(\omega t + \phi) \Leftrightarrow \mathbf{V} = V_m / \phi$$

(Time-domain representation) (Phasor-domain representation)

(b)

$$\begin{aligned} i &= -4 \sin(10t + 10^\circ) \text{ A} \\ &= 4 \cos(10t + 10^\circ + 90^\circ) \\ &= 4 \cos(10t + 100^\circ) \\ I &= 4 \angle 100^\circ \end{aligned}$$

remains, it is easy to show that

$$\begin{aligned} \sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \\ \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t \end{aligned}$$

✓ Find the sinusoids corresponding to these phasors:

(a) $\mathbf{V} = -25 \angle 40^\circ \text{ V}$

(b) $\mathbf{I} = j(12 - j5) \text{ A}$

phasor

domain

time
domain

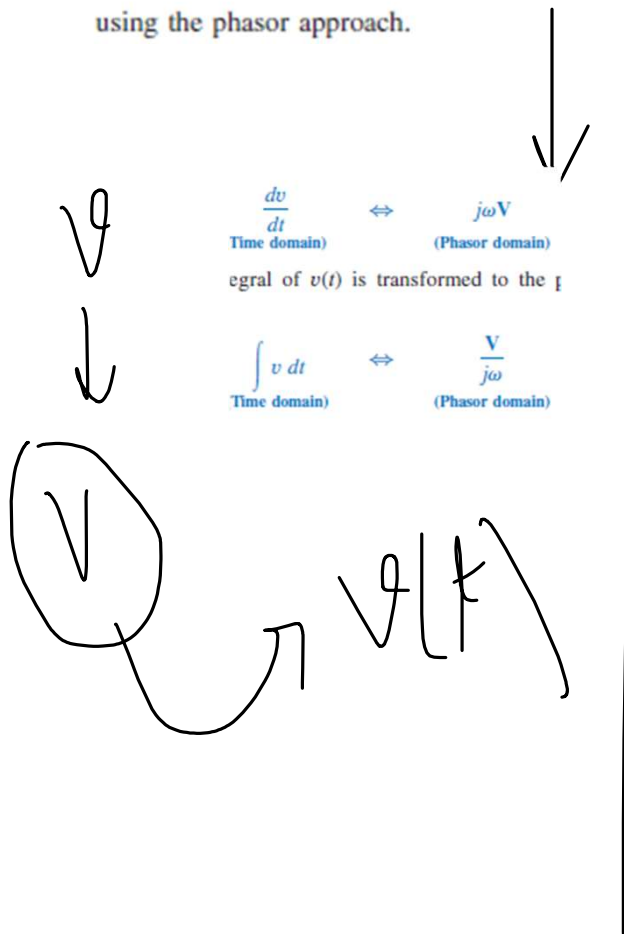
$$\begin{aligned} v &= -25 \cos(\omega t + 40^\circ) \\ &= 25 \cos(\omega t + 40^\circ - 180^\circ) \\ &= 25 \cos(\omega t - 140^\circ) \end{aligned}$$

$$\begin{aligned}
 \textcircled{A} \quad i &= jx(12 - j5) \\
 &= (11 \angle 90^\circ) \times (13 \angle -22.61^\circ) \\
 &= 13 \angle 67.39^\circ \\
 \therefore i &= 13 \cos \omega t + \underline{67.39^\circ}
 \end{aligned}
 \quad \bigg| \quad j = (1 \angle 90^\circ)$$

Find the voltage $v(t)$ in a circuit described by the integrodifferential equation

$$2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ)$$

using the phasor approach.



$$\begin{aligned}
 & 2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ) \\
 & \Rightarrow 2 j\omega V + 5V + 10 \times \frac{1}{j\omega} \times V = 50 \angle -30^\circ \\
 & \Rightarrow V \left(2j\omega + 5 + \frac{10}{j\omega} \right) = 50 \angle -30^\circ
 \end{aligned}$$

$$V = \frac{5 \angle -30^\circ}{\left(2j\omega^5 - j\frac{10}{\omega^5} \right)}$$

~~0~~ $\omega \rightarrow$ from the sinusoid.

$$, \therefore \omega = 5$$

$$V = 5.299 \angle -87.99^\circ$$

$$v(t) = 5.3 \cos(\omega t - 88^\circ)$$

PC (EEE101)

time domain

R

C

~~I~~

L



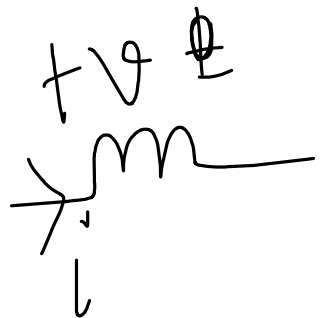
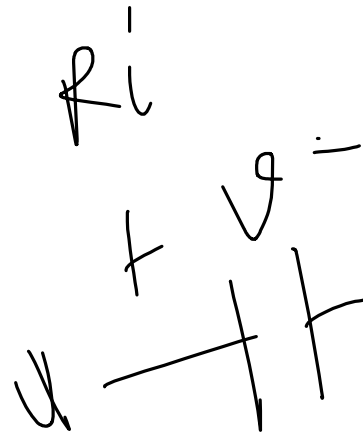
$$R = \frac{V}{i} \Rightarrow V = Ri$$



$$i = C \frac{dV}{dt}$$



$$V = L \frac{di}{dt}$$



$v(t)$ $i(t)$

$$R \rightarrow V = RI$$

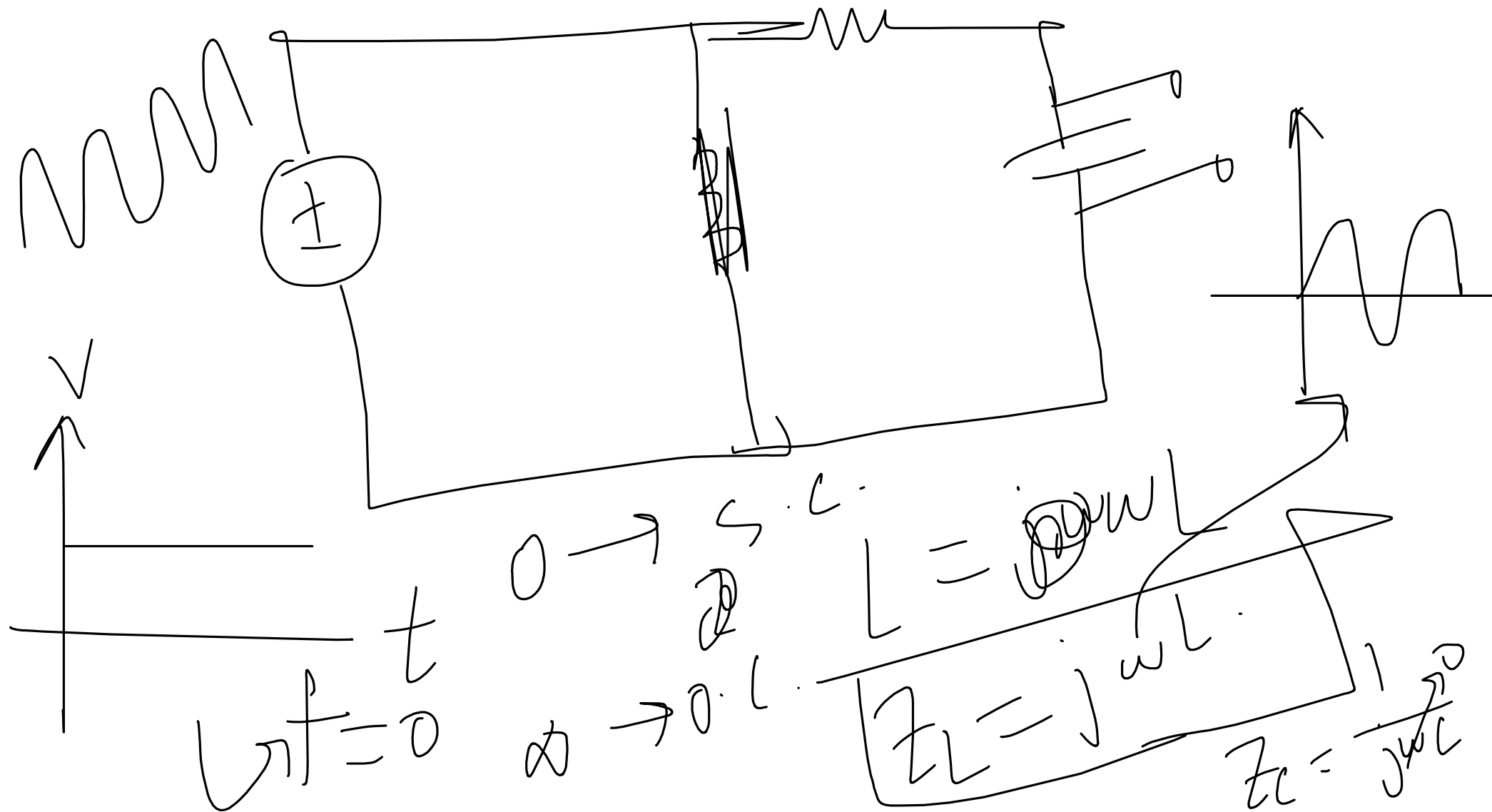
$$L \rightarrow V = L \frac{di}{dt}$$

$$C \rightarrow I = C \frac{dv}{dt} \quad \therefore V = \frac{I}{C \frac{dv}{dt}}$$

Impedance

$$Z = \frac{V}{I}$$

both in phasor form



Admittance

$$R \rightarrow \frac{1}{R} \leftarrow G \checkmark$$

$$\underline{Z} \rightarrow \frac{1}{Z} = Y$$

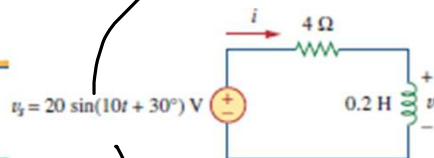
$$Z = R + jX \rightarrow \text{Reactance}$$

$$Y = G + jB \rightarrow \text{Susceptance}$$

Refer to Fig. 9.17. Determine $v(t)$ and $i(t)$.

Answer: $8.944 \sin(10t + 93.43^\circ)$ V, $4.472 \sin(10t + 3.43^\circ)$ A.

Practice Problem 9.9



5 | 10

remains, it is easy to show that

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$V_s =$$

$$V_s = 20 \sin(10t + 30^\circ)$$

$$= 20 \cos(10t + 30^\circ - 90^\circ)$$

$$= 20 \cos(10t - 60^\circ)$$

$$V_s = 20 \angle -60^\circ$$

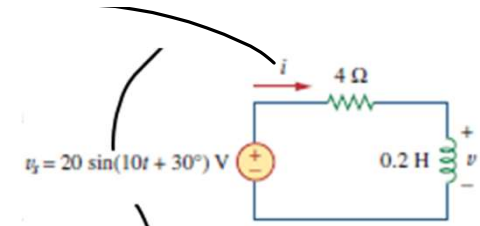
$$\begin{aligned}
 Z &= 4 + j\omega L \\
 &= 4 + j(10 \times 0.2) \\
 &= 4 + j2
 \end{aligned}$$

$$V_s = I \times Z$$

$$\therefore I = \frac{V_s}{Z}$$

$$= \frac{20 \angle -60^\circ}{4 + j2}$$

$$= 4.47 \angle -86^\circ$$



$$v = i \times Z$$

$$i(t) = 4.47 \cos(10t - 86^\circ)$$

$$\begin{aligned}
 V &= I \times Z_L \\
 &= \frac{I}{\sqrt{2}} \times j\omega L \\
 &= 4.47 \angle -86^\circ \times 2j
 \end{aligned}$$

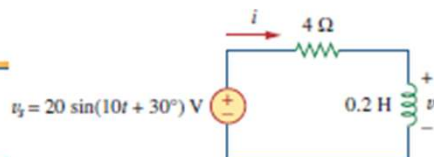
$$\therefore V = 8.89 \angle 3.43$$

$$V(f) = 8.89 \angle \cos(10t + 3.34)$$

Refer to Fig. 9.17. Determine $v(t)$ and $i(t)$.

Practice Problem 9.9

Answer: $8.944 \sin(10t + 93.43^\circ)$ V, $4.472 \sin(10t + 3.43^\circ)$ A.



$$V_s = 20 \sin(10t + 30^\circ)$$

$$V_s = 20 \angle 30^\circ$$

$$I = \frac{V_s}{Z}$$

$$= \frac{20 \angle 30^\circ}{4 + j2}$$

$$= 4.472 \angle 3.43^\circ$$

$$V = I \times Z_L$$

$$= I \times j\omega L$$

$$= 4.47 \angle 3.43^\circ \times j \times 10 \times 0.2$$

$$= 8.94 \angle 93.43^\circ$$